Heston model: 2D PDE pricing via Alternating Direction Implicit (ADI) schemes

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Contents

1 Introduction and goals **2**

2 The Heston PDE for European options **3**
2.1 Derivation via Feynman-Kac . . . 3
2.2 Change of variables: log-spot . . . 3
2.3 The reaction term and the discount factor trick . . . 3
2.4 Time reversal . . . 4

3 Computational domain and boundary conditions **4**
3.1 Truncation of the unbounded domain . . . 4
3.2 Boundary conditions in X . . . 4
3.3 Boundary condition at $v = 0$ . . . 4
3.4 Boundary condition at $v = v_{\max}$ . . . 5
3.5 Initial condition . . . 5

4 Spatial discretisation **5**
4.1 Grids . . . 5
4.2 Finite difference operators . . . 6
4.3 Operator decomposition . . . 6

5 ADI schemes **6**
5.1 The challenge . . . 6
5.2 Douglas scheme . . . 6
5.3 Craig-Sneyd (CS) scheme . . . 7
5.4 Hundsdorfer-Verwer (HV) scheme . . . 7
5.5 Implementation choice . . . 7

6 Stability **8**
6.1 Why explicit fails . . . 8
6.2 Why ADI works . . . 8

7 Convergence and validation **8**
7.1 Expected convergence rates . . . 8
7.2 Validation strategy . . . 9

8 Implementation outline **9**
8.1 Python 9
8.2 C++ 9

9 Bibliographical notes **10**

1 Introduction and goals

The preceding Heston notes give three independent pricing methods for European options under Heston: Fourier inversion (semi-analytic, fast and accurate but limited to specific payoffs), full-truncation Euler MC (robust but biased), and Andersen QE MC (less biased but still statistical). This note adds a fourth: deterministic PDE pricing on a 2D grid in (S, v).

PDE pricing has structural advantages over Monte Carlo for this problem:

- Deterministic: no statistical error, only discretisation error in space and time. The error decreases predictably with grid refinement.
- Greeks for free: derivatives of the price with respect to underlyings (Δ, Γ, vega) are extracted directly from the grid by finite differences, with no extra simulation cost.
- Natural fit for early exercise: American-style payoffs are handled by adding a free-boundary projection (PSOR or similar) to the time-stepping. The 1D version of this is the American put via PSOR (see the note *Pricing the American put via PSOR*).

The cost is implementation complexity. The Heston PDE is two-dimensional, with a non-trivial cross-derivative term coupling the two spatial directions. A naive explicit time-stepping is severely constrained by stability (the CFL condition forces $\Delta t = O(\Delta x^2 \Delta v^2 / \cdots)$, which in two dimensions kills runtime). The remedy is implicit time-stepping, but a fully implicit 2D solver requires inverting a large sparse matrix at every step. The compromise that wins in practice is *Alternating Direction Implicit* (ADI): split the implicit operator into directional pieces, each tridiagonal, and solve the directions sequentially. The Thomas solver (see the note *The implicit BTCS scheme and the Thomas algorithm*) is the workhorse.

This note has four goals:

The Heston PDE. Derive the pricing PDE in (S, v) via Feynman-Kac, transform to log-spot (X, v) for grid efficiency, and analyse the boundary conditions including the delicate behaviour at $v = 0$.

ADI schemes. Present the Douglas, Craig-Sneyd, and Hundsdorfer-Verwer methods, with emphasis on how each handles the mixed derivative term that appears in Heston (and in any 2D parabolic PDE with correlation structure). We commit to Douglas in the implementation because it captures all the ADI ideas in their simplest form; CS and HV are upgrades available if precision demands it.

Stability and convergence. Brief von Neumann analysis to motivate the implicit treatment, plus the discrete convergence rates: $O(\Delta x^2 + \Delta v^2)$ in space, $O(\Delta t)$ for Douglas, $O(\Delta t^2)$ for CS/HV.

Validation. Cross-check against the Fourier reference (`quantlib.heston_fourier`). At the default grid ($N_X \times N_v \times N_\tau = 200 \times 100 \times 100$) the PDE matches Fourier to 2-3 decimals; at $400 \times 200 \times 400$ the error is about $2 \cdot 10^{-3}$.

2 The Heston PDE for European options

2.1 Derivation via Feynman-Kac

Let $V(t, S, v)$ denote the price at time t of a European-style derivative paying $f(S_T)$ at maturity T. By the Feynman-Kac theorem applied to the Heston dynamics under $\mathbb{Q}$ (the Heston model note, equations (1)–(3)), V satisfies the parabolic PDE

$$\partial_t V + rS\partial_S V + \kappa(\theta - v)\,\partial_v V + \tfrac{1}{2}vS^2\partial_S^2 V + \rho\sigma vS\,\partial_S\partial_v V + \tfrac{\sigma^2}{2}v\,\partial_v^2 V - rV = 0, \tag{1}$$

on the domain $(t, S, v) \in [0, T) \times (0, \infty) \times [0, \infty)$, with terminal condition $V(T, S, v) = f(S)$.

The structure is parabolic in t, with a 2D second-order operator in (S, v). The terms in this operator are:

- Pure second derivatives: $\frac{1}{2}vS^2\partial_S^2 V$ (variance-of-spot) and $\frac{\sigma^2}{2}v\,\partial_v^2 V$ (variance-of-variance).
- **Mixed derivative**: $\rho\sigma vS\,\partial_S\partial_v V$. This is the term that makes the problem genuinely 2D and that ADI schemes must handle with care.
- First derivatives: $rS\partial_S V$ and $\kappa(\theta - v)\,\partial_v V$.
- Reaction: $-rV$.

For $\rho = 0$ the cross term vanishes and the problem decouples into two 1D problems coupled only through the source. For $\rho \neq 0$ (the financially interesting case), the cross term is essential and is what motivates the careful ADI construction.

2.2 Change of variables: log-spot

The PDE (1) has coefficients that depend on S (specifically, S^2 in the diffusion and S in the drift), which makes uniform grids in S poorly suited for the operator. The fix is the log-spot transformation $X = \log S$, $V(t, S, v) = U(t, X, v)$ where $U(t, X, v) := V(t, e^X, v)$. By the chain rule:

$$\partial_S V = \tfrac{1}{S}\partial_X U, \qquad \partial_S^2 V = \tfrac{1}{S^2}(\partial_X^2 U - \partial_X U), \qquad \partial_S\partial_v V = \tfrac{1}{S}\partial_X\partial_v U.$$

Substituting into (1) and simplifying:

$$\partial_t U + (r - \tfrac{1}{2}v)\partial_X U + \kappa(\theta - v)\partial_v U + \tfrac{1}{2}v\,\partial_X^2 U + \rho\sigma v\,\partial_X\partial_v U + \tfrac{\sigma^2}{2}v\,\partial_v^2 U - rU = 0, \tag{2}$$

on $(t, X, v) \in [0, T) \times \mathbb{R} \times [0, \infty)$, with terminal condition $U(T, X, v) = f(e^X)$.

Now *the coefficients of the operator do not depend on X at all*; they depend only on v. This means the discrete operator in X has constant coefficients and can be applied identically at every v-level. Uniform grids in X are now natural.

2.3 The reaction term and the discount factor trick

The reaction term $-rU$ in (2) is a nuisance for ADI: it doesn't fit cleanly into either a pure X or pure v direction. The clean solution is to remove it via the substitution

$$\tilde{U}(t, X, v) := e^{r(T-t)}U(t, X, v),$$

which absorbs the discount into the function. Then $\partial_t\tilde{U} = e^{r(T-t)}(\partial_t U - rU)$, and the reaction term is gone:

$$\partial_t\tilde{U} + (r - \tfrac{1}{2}v)\partial_X\tilde{U} + \kappa(\theta - v)\partial_v\tilde{U} + \tfrac{1}{2}v\,\partial_X^2\tilde{U} + \rho\sigma v\,\partial_X\partial_v\tilde{U} + \tfrac{\sigma^2}{2}v\,\partial_v^2\tilde{U} = 0, \tag{3}$$

with terminal condition $\tilde{U}(T, X, v) = f(e^X)$. This is the form we will discretise. After solving for $\tilde{U}$, the original price is recovered as $V(t, e^X, v) = e^{-r(T-t)}\tilde{U}(t, X, v)$.

2.4 Time reversal

The PDE (3) is parabolic with terminal condition; for numerical purposes it's cleaner to reverse time, defining $\tau := T - t$ and $W(\tau, X, v) := \tilde{U}(T - \tau, X, v)$. Then $\partial_t = -\partial_\tau$, the equation is forward in τ:

$$\partial_\tau W = (r - \tfrac{1}{2}v)\partial_X W + \kappa(\theta - v)\partial_v W + \tfrac{1}{2}v\,\partial_X^2 W + \rho\sigma v\,\partial_X\partial_v W + \tfrac{\sigma^2}{2}v\,\partial_v^2 W, \tag{4}$$

on $(\tau, X, v) \in (0, T] \times \mathbb{R} \times [0, \infty)$, with *initial* condition $W(0, X, v) = f(e^X)$. This is the form we discretise: a forward parabolic problem with explicit initial data, identical in structure to the heat equation modulo coefficient values.

3 Computational domain and boundary conditions

3.1 Truncation of the unbounded domain

The mathematical domain $\mathbb{R} \times [0, \infty)$ must be truncated to a bounded box $[X_{\min}, X_{\max}] \times [0, v_{\max}]$ for numerical work. The truncation introduces *numerical* boundary conditions which approximate the true behaviour of the solution at infinity.

Practical choices:

- $X_{\min} = \log S_0 - 4\sqrt{v_{\max}T}$, $X_{\max} = \log S_0 + 4\sqrt{v_{\max}T}$. This places the spot grid roughly $\pm 4\sigma$-equivalents from the at-the-money region, which captures essentially all the option's nonlinearity. Tighter truncation introduces noticeable boundary errors near the strike.
- $v_{\max} = 5\theta$ or larger. The variance is mean-reverting to θ and rarely exceeds a few times θ, but the tail matters for option pricing. 5θ is conservative for typical equity calibrations; for high vol-of-vol regimes (σ large), enlarge.

3.2 Boundary conditions in X

At $X = X_{\min}$ (deep out-of-the-money for a call, deep in-the-money for a put), and $X = X_{\max}$ (the symmetric direction), the code does not discretise L_X at all: on the boundary rows $i = 0$ and $i = N_X$ the explicit step sets $L_X W = 0$ (and $L_{Xv} W = 0$), and the implicit X-solve has identity rows. The boundary values therefore evolve only through the v-operator,

$$\partial_\tau W\big|_{X=X_{\min}} = L_v W, \qquad\qquad \partial_\tau W\big|_{X=X_{\max}} = L_v W.$$

For a call this is accurate at $X_{\min}$, where the price and its X-derivatives tend to zero. It is not exact at $X_{\max}$: there the price tends to $S - Ke^{-r\tau}$, so $W \approx e^{X+r\tau} - K$ and

$$\partial_X^2 W = S\partial_S W + S^2\partial_S^2 W \approx e^{X+r\tau} \neq 0;$$

dropping L_X keeps the boundary value close to the payoff $e^{X_{\max}} - K$ instead of $e^{X_{\max}+r\tau} - K$. The error is acceptable because the boundary is far away: $X_{\max} - \log S_0 = 4\sqrt{v_{\max}T} = 4\sqrt{5\theta T} \approx 9\sqrt{\theta T}$, about nine standard deviations of $\log S_T$ at the long-run variance, so the boundary error has a negligible effect at $(\log S_0, v_0)$.

3.3 Boundary condition at $v = 0$

The boundary at $v = 0$ requires careful analysis. From the PDE (4), at $v = 0$ all the second derivatives in v and the cross derivative vanish (their coefficients are proportional to v), and the variance-of-spot term $\frac{1}{2}v\partial_X^2 W$ also vanishes. The PDE *at* $v = 0$ degenerates to:

$$\partial_\tau W\big|_{v=0} = r\partial_X W\big|_{v=0} + \kappa\theta\,\partial_v W\big|_{v=0}. \tag{5}$$

This is a first-order PDE in (τ, X, v) at the boundary $v = 0$. The drift $\kappa\theta$ in v is positive (assuming $\kappa, \theta > 0$), so the characteristics of $\partial_\tau W = \kappa\theta\,\partial_v W$, along which $dv/d\tau = -\kappa\theta < 0$, run from the interior towards $v = 0$ and leave the domain there: the boundary $v = 0$ is *outflow*, and information flows from the interior to $v = 0$. By the theory of first-order PDEs, no boundary condition is needed at an outflow boundary; the equation (5) *is* the boundary condition, and we discretise it directly using upwind differencing in v.

This is a feature, not a bug: the variance process tends to mean-revert away from zero, and the PDE captures this without requiring us to specify an additional boundary value. Compare this with the more delicate treatment required for, e.g., the SABR PDE.

3.4 Boundary condition at $v = v_{\max}$

At $v = v_{\max}$ (truncation of the unbounded v-direction), the code drops the second derivative in v and the cross term, keeps L_X, and discretises the drift with a backward difference:

$$\partial_\tau W_{i,N_v} = (L_X W)_{i,N_v} + \kappa(\theta - v_{\max})\,\frac{W_{i,N_v} - W_{i,N_v-1}}{\Delta v}.$$

Dropping $\partial_v^2 W$ and $\partial_X \partial_v W$ is consistent with the large-variance limit: the call price is bounded by S and tends to S as $v \to \infty$, so W flattens in v and its v-derivatives tend to zero. The drift $\kappa(\theta - v_{\max})$ is negative since $v_{\max} > \theta$ in our setup, so the characteristics ($dv/d\tau = \kappa(v_{\max} - \theta) > 0$) carry information from the interior to the boundary; the backward difference is the upwind one and uses only interior values, so no boundary value has to be prescribed.

3.5 Initial condition

For a European call with strike K, the terminal payoff is $f(S) = \max(S - K, 0)$, so

$$W(0, X, v) = \max(e^X - K, 0).$$

This is independent of v, which is intuitive: at maturity, only the spot matters. This initial condition has a non-smooth corner at $X = \log K$ (the strike), which propagates a known oscillation in the numerical solution unless smoothed. Here we use the unsmoothed payoff and let the smoothing happen via the discrete operator's diffusion; in production calibration one would replace it with a C^2 smoothed version near the strike.

4 Spatial discretisation

4.1 Grids

Uniform grids in both directions:

$$\begin{aligned} X_i &= X_{\min} + i\,\Delta X, \quad i = 0, \ldots, N_X, \quad \Delta X = (X_{\max} - X_{\min})/N_X, \\ v_j &= j\,\Delta v, \qquad\qquad j = 0, \ldots, N_v, \quad \Delta v = v_{\max}/N_v. \end{aligned}$$

The grid has $(N_X + 1)(N_v + 1)$ nodes. We denote the discrete approximation $W_{ij}^n \approx W(\tau_n, X_i, v_j)$, where $\tau_n = n\Delta\tau$.

The implementation uses uniform grids in both directions. A non-uniform grid in v (concentrated near v_0 and near $v = 0$) would give better accuracy at fixed N_v, since the option price is most sensitive to variance values near v_0; it is not implemented here.

4.2 Finite difference operators

For the second derivatives we use centred differences (standard $O(\Delta^2)$ accuracy):

$$\partial_X^2 W\big|_{ij} \approx \frac{W_{i+1,j} - 2W_{ij} + W_{i-1,j}}{\Delta X^2}, \qquad \partial_v^2 W\big|_{ij} \approx \frac{W_{i,j+1} - 2W_{ij} + W_{i,j-1}}{\Delta v^2}.$$

For the first derivatives we again use centred differences in the interior, but switch to upwind one-sided differences at $v = 0$ and $v = v_{\max}$ to maintain stability (no X-stencil is applied at the X-boundaries).

The cross-derivative is approximated by:

$$\partial_X \partial_v W\big|_{ij} \approx \frac{W_{i+1,j+1} - W_{i+1,j-1} - W_{i-1,j+1} + W_{i-1,j-1}}{4\Delta X \Delta v},$$

which is centred and $O(\Delta^2)$. This is a 9-point stencil in (X, v).

4.3 Operator decomposition

Following In 't Hout & Foulon (2010), we decompose the discrete operator L into three parts:

$$L = L_X + L_v + L_{Xv}. \tag{6}$$

where:

- L_X contains all derivatives in X only (and their coefficients): $(r - \frac{1}{2}v)\partial_X + \frac{1}{2}v\partial_X^2$.
- L_v contains all derivatives in v only: $\kappa(\theta - v)\partial_v + \frac{\sigma^2}{2} v\, \partial_v^2$.
- L_{Xv} is the cross term: $\rho\sigma v\, \partial_X \partial_v$.

This decomposition is the basis for ADI schemes: L_X and L_v are tridiagonal in their respective directions when written as matrices, while L_{Xv} couples both directions but is treated explicitly.

5 ADI schemes

5.1 The challenge

A fully implicit time step from W^n to W^{n+1} would solve $W^{n+1} - \Delta\tau\, L W^{n+1} = W^n$, where L is the full 2D operator. The matrix $I - \Delta\tau L$ is sparse but *not* tridiagonal in any direction; inverting it directly costs $O((N_X N_v)^{1.5})$ or more.

ADI splits the implicit step into directional pieces, each of which is tridiagonal and solvable by Thomas in $O(N_X N_v)$. The trick is how to handle L_{Xv}, which couples both directions.

5.2 Douglas scheme

The Douglas scheme (also called "Fractional Step" or "Predictor-Corrector ADI") is the simplest. One step from τ_n to τ_{n+1}:

1. **Predictor (explicit Euler with full operator)**:

$$Y_0 = W^n + \Delta\tau\, L\, W^n.$$

2. **Implicit correction in X**:

$$Y_1 - \theta\Delta\tau\, L_X(Y_1 - W^n) = Y_0,$$

or equivalently $Y_1 = Y_0 + \theta\Delta\tau\, L_X(Y_1 - W^n)$. This is solvable as a tridiagonal system in X at each j.

3. **Implicit correction in v**:

$$Y_2 - \theta\Delta\tau\, L_v(Y_2 - W^n) = Y_1.$$

Tridiagonal in v at each i.

4. Set $W^{n+1} = Y_2$.

Here θ is the implicitness parameter; $\theta = 1/2$ gives Crank-Nicolson-like behaviour and $\theta = 1$ is fully implicit. For Heston, $\theta = 1/2$ is the standard choice.

The cross term L_{Xv} enters only the explicit predictor stage; it is never inverted. This is the central insight of Douglas ADI: *the directional pieces are inverted; the cross piece is treated explicitly.*

Order. Douglas is first-order accurate in τ, $O(\Delta\tau)$, regardless of θ. The explicit treatment of L_{Xv} is what limits the temporal order; if all three pieces were implicit (fully implicit ADI), the order would be $O(\Delta\tau^2)$ but at much higher cost.

5.3 Craig-Sneyd (CS) scheme

The Craig-Sneyd scheme adds a second-order correction. After computing Y_2 as in Douglas, one performs additional explicit and implicit correction stages that combine to give $O(\Delta\tau^2)$ accuracy. The full CS scheme has 5 stages per time step (vs Douglas's 3), and is approximately twice as expensive per step, but converges with second-order rate.

5.4 Hundsdorfer-Verwer (HV) scheme

HV is a further refinement of CS that improves stability for stiff problems. For Heston, In 't Hout & Foulon recommend first the modified Craig-Sneyd (MCS) scheme with $\theta = 1/3$, with CS ($\theta = 1/2$) and HV ($\theta = \frac{1}{2} + \frac{\sqrt{3}}{6}$) as good alternatives. Like CS, HV is $O(\Delta\tau^2)$, with similar cost structure; the difference is in the placement of the θ parameter and the structure of the corrector stages, which give HV better stability properties at large $\Delta\tau$.

5.5 Implementation choice

We implement **Douglas** as the production scheme. Reasons:

- It illustrates all the ADI ideas (operator splitting, tridiagonal sweeps, explicit cross term) in their simplest form.
- $O(\Delta\tau)$ in time is acceptable when paired with reasonable space discretisation: at the default grid sizes ($N_X \times N_v \times N_\tau = 200 \times 100 \times 100$), the Douglas error is dominated by the spatial $O(\Delta x^2 + \Delta v^2)$ rather than the temporal $O(\Delta\tau)$.
- Upgrading to CS or HV is mechanical: the discrete operators L_X, L_v, L_{Xv} are identical; only the time-stepping orchestration changes.

If higher precision is required (e.g., for option Greeks where $O(\Delta\tau)$ truncation is amplified), the Hundsdorfer–Verwer scheme is the appropriate upgrade; it is not implemented here.

6 Stability

A full von Neumann stability analysis of the Heston ADI is complex and depends on the parameter regime. We give the qualitative conclusions.

6.1 Why explicit fails

A fully explicit scheme $W^{n+1} = W^n + \Delta\tau\, LW^n$ is constrained by the CFL condition. For 2D parabolic problems with diffusion coefficients D_x, D_v, the constraint is

$$\Delta\tau \leq \frac{1}{2(D_x/\Delta X^2 + D_v/\Delta v^2)}.$$

For Heston, $D_x = \frac{1}{2}v$ and $D_v = \frac{\sigma^2}{2}v$, both of which depend on v. At $v = v_{\max} = 5\theta$, with typical parameters ($\theta = 0.04$, $\sigma = 0.3$, $T = 1$) and $\Delta X = \Delta v = 0.05$, this gives $\Delta\tau \lesssim 1.1 \cdot 10^{-2}$. The grids actually used are much finer in v: with $\Delta X = 8\sqrt{v_{\max}T}/N_X$ and $\Delta v = v_{\max}/N_v$, the 100×50 grid ($\Delta X \approx 0.036$, $\Delta v = 0.004$) gives $\Delta\tau \lesssim 7.8 \cdot 10^{-4}$, about 1,300 time steps for $T = 1$, and the default 200×100 grid ($\Delta X \approx 0.018$, $\Delta v = 0.002$) gives $\Delta\tau \lesssim 2.0 \cdot 10^{-4}$, about 5,100 steps. ADI lets us use $\Delta\tau \sim 10^{-2}$, roughly 10 to 50 times fewer steps.

6.2 Why ADI works

The directional implicit treatments of L_X and L_v remove the CFL constraints from those directions: the resulting tridiagonal solves are stable at any $\Delta\tau$. The cross term L_{Xv} is treated explicitly, and it is not small: on the 100×50 grid above, $|\rho|\sigma v_{\max}/(\Delta X \Delta v) \approx 294$ (with $\rho = -0.7$) against $D_x/\Delta X^2 \approx 78$ for the X-diffusion (both at $v = v_{\max}$), and both grow by a factor 4 on the 200×100 grid. It does not bring back a step-size restriction, however: In 't Hout and Welfert [6] prove that the Douglas scheme with $\theta \geq 1/2$ is unconditionally stable, in the von Neumann sense (constant coefficients, periodic boundary conditions), for 2D convection-diffusion equations with a mixed derivative term (see In 't Hout and Foulon [5] for the Heston case). In our runs Douglas with $\theta = 1/2$ gives bounded prices on the default 200×100 grid with only $N_\tau = 5$ or 10 steps; at such step sizes the error is the temporal discretisation error, not an instability.

In practice, $\Delta\tau = T/100$ to $T/200$ is reliable for Douglas with typical Heston parameters and $N_X, N_v \sim 100$.

7 Convergence and validation

7.1 Expected convergence rates

- Spatial: $O(\Delta X^2 + \Delta v^2)$ with centred differences.
- Temporal (Douglas): $O(\Delta\tau)$.
- At the default grid sizes ($N_X = 200, N_v = 100, N_\tau = 100$), the spatial error dominates. Refining N_X, N_v by factor 2 each should reduce the error by factor 4.

7.2 Validation strategy

The Fourier reference is the ground truth: it is bias-free to numerical quadrature precision ($\sim 10^{-6}$). The PDE pricer matches it up to its discretisation error, which is much larger: 2-3 decimals at the default grid, and about $2 \cdot 10^{-3}$ at $400 \times 200 \times 400$ (`pde_adi_convergence.csv` in `python/results/phase4`).

1. **BS limit ($\sigma \to 0$).** As for the Monte Carlo pricers, the Heston PDE reduces to BS in this limit; verify the pricer matches BS analytic.

2. **Cross-method vs Fourier.** For a panel of strikes and maturities, the PDE price agrees with Fourier to 2-3 decimals on the default 200×100 spatial grid (errors up to $8 \cdot 10^{-3}$, largest at the money: Test 2 of `validate_heston_pde.py` and `heston_cross_validation.csv`), and to about $2 \cdot 10^{-3}$ at $400 \times 200 \times 400$.

3. **Convergence study.** Halve $\Delta X, \Delta v, \Delta\tau$ and observe the error decrease following the expected rates.

4. **Cross-method vs MC (sanity).** The QE pricer (`quantlib.heston_qe`) with large M provides a third independent estimate. PDE and QE should agree within MC half-widths.

8 Implementation outline

8.1 Python

The module `quantlib.heston_pde` has one public function, built on private helpers:

- `heston_call_pde(S0, K, T, kappa, theta, sigma, rho, v0, r, N_X=200, N_v=100, N_tau=100, *, theta_imp=0.5, X_factor=4.0, v_max_factor=5.0)`: high-level pricer. It builds the uniform (X, v) grids itself (the X-grid extends $X_{\text{factor}} \cdot \sqrt{v_{\max} T}$ either side of $\log S_0$, the v-grid extends to $v_{\max} = v_{\max_\text{factor}} \cdot \theta$), precomputes the tridiagonal coefficients of L_X and L_v (`_build_LX_coefficients`, `_build_Lv_coefficients`), sets the call payoff initial condition, runs N_τ Douglas steps, interpolates bilinearly at $(\log S_0, v_0)$ and discounts by e^{-rT}.

- `_douglas_step(W, dtau, dX, dv, v_grid, kappa, theta, sigma, rho, theta_imp, LX_a, LX_b, LX_c, Lv_a, Lv_b, Lv_c)`: one Douglas step from the precomputed coefficients; it does not build the operators. It applies L_X, L_v, L_{Xv} explicitly for the predictor and runs the two tridiagonal solves (one in X, one in v).

The module has its own batched Thomas solver (`_thomas_batch`): in the ADI sweeps each line of the grid produces a tridiagonal system, and the systems of all lines are solved together in vectorised form.

8.2 C++

The C++ implementation in `cpp/include/quant/heston_pde.hpp` and `cpp/src/heston_pde.cpp` mirrors Python with the standard adaptations: `HestonParams` from `heston_fourier.hpp`, its own batched Thomas solver, `std::vector<double>` for the flattened 2D grid. The 2D grid is stored row-major (X-major: index $(i, j) \to i \cdot (N_v + 1) + j$).

The Catch2 tests in `cpp/tests/test_heston_pde.cpp` verify the four pillars at reduced grid sizes for runtime, with snapshot agreement against Python and Fourier.

9 Bibliographical notes

- In 't Hout, Foulon (2010) [5]: the standard reference for ADI applied to Heston. Detailed comparison of Douglas, CS, MCS, HV; recommends MCS with $\theta = 1/3$, with CS ($\theta = 1/2$) and HV ($\theta = \frac{1}{2} + \frac{\sqrt{3}}{6}$) as good alternatives.
- In 't Hout, Welfert (2007) [6]: von Neumann stability of the Douglas and CS schemes for convection-diffusion equations with mixed derivative terms; unconditional stability for $\theta \geq 1/2$.
- Hundsdorfer, Verwer (2003) [4]: classical text on ADI and IMEX methods, with the original HV scheme.
- Peaceman, Rachford (1955) [7]: the original ADI paper.
- Douglas, Rachford (1956) [3]: ADI for heat conduction in two and three space variables, now of historical interest.
- Craig, Sneyd (1988) [2]: the second-order correction to Douglas.
- Achdou, Pironneau (2005) [1]: general PDE methods in finance, includes Heston PDE.

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